DO NOW — RECALL

Lesson 070 · Heuristics and Performance Modelling

Name: _____ Class: _____ Date: _____

1. What is correlation?

2. What does a heuristic provide?

3. What is an optimisation objective?

4. A sorted 32-element candidate set is halved to one candidate. How many halvings are needed?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Statistical association, not proof of causation.

2. Model answer

An estimate guiding an algorithm's next choice.

3. Model answer

A quantity to minimise/maximise.

4. Model answer

Five: 32→16→8→4→2→1.

By the end, you can…

• A heuristic is a useful estimate or rule of thumb that guides decisions when complete exploration is expensive.

• A locally cheapest edge does not necessarily lead to a globally cheapest route; evaluate candidate routes as complete paths.

• Performance modelling estimates time, memory or cost under assumptions. A predicted result requires checking against real hardware and workloads.

Vocabulary — know these terms

heuristic — Estimate guiding solution search.

greedy — Choose apparently best immediate option.

performance model — Simplified resource/time estimate.

admissible — Heuristic never overestimates remaining optimal cost.

NEW LEARNING

Read, annotate and explain

Define and predict

A heuristic is a useful estimate or rule of thumb that guides decisions when complete exploration is expensive.

Explain

A locally cheapest edge does not necessarily lead to a globally cheapest route; evaluate candidate routes as complete paths.

Evaluate

Performance modelling estimates time, memory or cost under assumptions. A predicted result requires checking against real hardware and workloads.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Choose cheaper complete path: X→Y→G is 3+9; X→Z→G is 6+2.

Follow the reasoning

Calculate the entire feasible route rather than the next immediate edge.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Compare route A-B-G edges 2+8 against A-C-G 5+1. Calculate totals and select cheaper. [6]

Question 2

Model 1200 jobs processed at 80/s, under constant throughput: estimate duration and one assumption. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate using straight-line distance as heuristic for car routes with one-way roads and congestion. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A local greedy choice always produces an optimal complete path.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.